PRODUCT USER MANUAL

For

Black Sea Ocean Heat Content Anomaly (0-300m)

OMI_CLIMATE_OHC_BLKSEA_area_averaged_anomalies

Issue 2.0

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Change Record

Issue	Date	§	Description of Change	Author	Checked by	Validated by
1.0	2022/11/29	all	Creation of the document	L. Lima, R. Lecci		
2.0	2025/05/28	all	Updated time series based on BLK-REA BSe4r1 and setup of operational extension	L. Lima, R. G de Menezes	R. Lecci	Copernicus Marine Product Management

GLOSSARY AND ABBREVIATIONS

MOI	Mercator Océan International
NetCDF	Network Common Data Form
OMI	Ocean Monitoring Indicator
PUM	Product User Manual
QUID	Quality Information Document

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DATA ACCESS

After registration, you will be able to download our data. To assist you, our HelpCenter is available, and more specifically its section about download.

Information on operational issues on products and services can be found on our User Notification Service. If you have any questions, please contact us.

I INTRODUCTION

I.1. Summary

This document is the user manual for the CMEMS Ocean Monitoring Indicator (OMI) for the Black Sea : OMI_CLIMATE_OHC_BLKSEA_area_averaged_anomalies. It provides a 1D (time dimension) monthly mean time series of Ocean Heat Content (OHC) area averaged anomalies calculated for the Black Sea between 0 and 300 meters depths excluding data with topography lower than 300 meters (see QUID document for more information on the calculation).

It is computed from different reanalysis and observational datasets. The dataset contains monthly mean time series of OHC averaged over the Black Sea. This OMI is computed using the regional Black Sea reanalysis (BLK-REA), the global reanalysis (GLORYS12V1), and CORA5.2, ARMOR3D, which are temperature objective analyses datasets prepared using observations.

The datasets above are used according to their time extension. Time series of the BLK-REA and its results in Interim mode are used covering up to the year before the current one [-1Y]. The ensemble mean is calculated over the time period common to all datasets, starting from 2005 up to three years preceding the current year [-3Y]. This OMI is operationally extended on a yearly basis.

The product is published on the CMEMS dissemination server after automatic and delayed-mode quality controls. The product is available online and disseminated through the CMEMS Information System. Files downloaded are in NetCDF format and follow CF-1.4 convention.

Disclaimer: The quality of the product may vary during the proposed time series depending on the possible update of the system.

Further details are described in the Quality Information Document (QUID):

- CMEMS-BLK-OMI-QUID-CLIMATE- OHC V3.0 .pdf: https://doi.org/10.48670/moi-00306

See also News flash. More detailed information can be obtained from http://marine.copernicus.eu/services-portfolio/contact-us/

Information on operational issues on products and services can be found on our **User Notification Service**. If you have any questions, please **contact us**.

I.2. History of changes

2022/11	1st publication of the product: OMI_CLIMATE_OHC_BLKSEA_area_averaged_anomalies
2025/11	Update the time series based on the results from BLK-REA version BSe4r1 and according to the availability of datasets Setup of operational extensions

II DESCRIPTION OF THE PRODUCT

II.1. General information

Product(s) name	OMI_CLIMATE_OHC_BLKSEA_area_averaged_anomalies
Geographical coverage	Black Sea
Variables	Ocean Heat Content (OHC)
Update frequency	Yearly
Available time series	From 2005 until -3 years before the current one
Delivery mechanism	MDS
Temporal/Spatial resolution	monthly values
Integration depth (for example..)	0-300m
Format	NetCDF CF1.7

II.2. Details of the dataset

omi_climate_ohc_blksea_area_averaged_anomalies contains the 1D annual Black Sea OHC anomaly from: BLKREA, GLORYS12V1, ARMOR and CORA5.2 observations and the ensemble mean OHC as well as the corresponding standard deviation.
ohc_mean [J.m-2] long_name: Ensemble Mean of Area Averaged Ocean Heat Content Anomaly standard_name: integral_wrt_depth_of_sea_water_potential_temperature_expressed_as_heat_content **ohc_std** [J.m-2] long_name: Ensemble Standard Deviation of Area Averaged Ocean Heat Content Anomaly standard_name: integral_wrt_depth_of_sea_water_potential_temperature_expressed_as_heat_content **ohc_glor** [J.m-2] long_name: Area Averaged Ocean Heat Content Anomaly from GLORYS12V1 Sea Water Potential Temperature standard_name: integral_wrt_depth_of_sea_water_potential_temperature_expressed_as_heat_content **ohc_armor** [J.m-2] long_name: Area Averaged Ocean Heat Content Anomaly from ARMOR-3D Sea Water Temperature standard_name: integral_wrt_depth_of_sea_water_temperature_expressed_as_heat_content **ohc_cora** [J.m-2] long_name: Area Averaged Ocean Heat Content Anomaly from CORA5.2 Sea Water Temperature standard_name: integral_wrt_depth_of_sea_water_temperature_expressed_as_heat_content **ohc_bs** [J.m-2] long_name: Area Averaged Ocean Heat Content Anomaly from Black Sea Water Potential Temperature standard_name: integral_wrt_depth_of_sea_water_potential_temperature_expressed_as_heat_content

II.3. Product System Description

Data are extracted from the Black Sea regional reanalysis (BS-REA), global reanalysis (GLORYS 12V1), and two objective analyses products: ARMOR3D and CORA5.2. These datasets provide 3D temperature fields used to compute monthly Ocean Heat Content (OHC) anomalies, averaged over the upper 300 meters of the water column. The resulting time series, derived from multiple sources, cover the Black Sea region. The monthly mean time series start from 2005 up to three years prior to the current year [-3Y] .

III FILES NOMENCLATURE AND FORMAT

Information about nomenclature of files when downloaded can be found in this article: " How is defined the nomenclature of Copernicus Marine data? | Copernicus Marine Help Center "

III.1. Nomenclature of files

The files nomenclature when downloaded through the Copernicus Marine Service Download Services is based on the following nomenclature:

<datasetname> <.nc >

Where:

- **<datasetname>** is the dataset reference**.**
- **.nc** is the standard NetCDF filename extension.

For example: **omi_climate_ohc_blksea_area_averaged_anomalies.nc**

III.2. File Format

- ✓ The products are stored using the NetCDF format.
- ✓ To know more about the NetCDF format, please follow this link:
- ✓ What is the format of Copernicus Marine products ? NetCDF
- ✓ To understand the differences between netCDF and Zarr, please consult this article:
- ✓ how-to-choose-between-netcdf-and-zarr-format-using-the-toolbox

IV REFERENCES

See the Quality Information Document (QUID CMEMS-OMI-QUID-CLIMATE-BLKSEA-OHC.pdf, https://doi.org/10.48670/moi-00306